

POWER BI MULTI-SOURCE INTEGRATION & DASHBOARD PROJECT

Retail Sales & Inventory Analytics Dashboard

Business Domain: Retail Sales & Inventory

Tool Used: Microsoft Power BI Desktop

Data Sources: 4 (Excel, CSV, JSON, CSV)

Total Records: 7,416+ Rows

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1. Introduction

This project covers the design and development of a comprehensive Business Intelligence (BI) dashboard built with Microsoft Power BI Desktop. It focuses on the Retail Sales and Inventory domain, simulating a real-world BI consulting scenario. Data from four different sources has been brought together, cleaned, modeled, and turned into interactive visualizations that support data-driven decision-making.

Microsoft Power BI is one of the most widely used business analytics platforms today. In this project, it was used to connect multiple data sources, build a star schema data model, write DAX measures, and design a professional multi-page interactive dashboard from start to finish.

2. Business Problem

The retail company at the center of this project was struggling with several operational pain points that made it difficult to get a clear picture of business performance:

- No centralized reporting system across multiple store locations in different cities
- Difficulty tracking sales performance, profitability, and inventory levels in real time
- Limited visibility into customer segments and their purchasing patterns
- Slow, manual reporting processes with no forecasting capability in place
- No way to benchmark actual sales against monthly and annual targets

This Power BI dashboard addresses all of these gaps by pulling data from multiple sources into a single, unified reporting platform. Decision-makers can now track KPIs, explore trends, drill into the data, and make better business decisions without relying on spreadsheets or manual reports.

3. Data Sources Used

The project connects four distinct data sources to Power BI, as required by the project specifications. The table below summarizes each source:

#	File Name	Source Type	Records	Tables	Description
1	products.xlsx	Excel Workbook	320 rows	2 sheets	Product catalog with categories, pricing, and stock levels
2	customers.csv	CSV File	2,000 rows	1 table	Customer info including city, region, and segment
3	sales_transactions.csv	CSV File	5,000 rows	1 table	Sales records with profit, payment method, and channel

#	File Name	Source Type	Records	Tables	Description
4	store_targets.json	JSON File	8 stores	1 table	Monthly and annual sales targets per store

Total combined rows across all sources: 7,416+ rows — comfortably exceeding the minimum requirement of 5,000 rows.

4. Data Cleaning & Transformation (Power Query ETL)

All data transformation was handled inside Power Query Editor in Power BI. The following ETL (Extract, Transform, Load) steps were applied to each table:

4.1 Sales Transactions Table

- SaleDate column converted to Date data type
- TotalSales, Profit, UnitPrice, and UnitCost columns converted to Decimal Number
- Duplicate rows removed using Remove Duplicates
- Null values in the ProfitMargin column replaced with 0
- Custom calculated column added: SaleMonth_Year for time-based analysis

4.2 Customers Table

- JoinDate column converted to Date data type
- Age column converted to Whole Number
- Duplicate customer records removed
- Custom column added: AgeGroup (Youth, Adult, Middle Age, Senior)

4.3 Products Table

- LaunchDate converted to Date data type
- CostPrice and SellingPrice converted to Decimal Number
- Calculated columns added: ProfitPerUnit and Margin_Percentage

4.4 Date Table (Created from Scratch)

- Custom DateTable built using Power Query M Language
- Covers January 2023 through December 2024
- Additional columns added: Year, Month, MonthName, Quarter, WeekDay, MonthShort
- This table drives all time intelligence DAX calculations throughout the model

5. Data Model

The data model follows a Star Schema design, which is the industry standard for BI reporting. The Sales Transactions table is the central Fact Table, while Products, Customers, and DateTable act as Dimension Tables surrounding it.

5.1 Star Schema Structure

From Table	Column	To Table	Relationship Type
sales_transactions	ProductID	Products	Many to One (*)
sales_transactions	CustomerID	Customers	Many to One (*)
sales_transactions	SaleDate	DateTable	Many to One (*)

All relationships use a single cross-filter direction, which keeps the model clean and query performance fast.

5.2 DAX Measures

The following DAX (Data Analysis Expressions) measures were created to calculate KPIs and business metrics dynamically:

Measure Name	DAX Formula	Purpose
Total Sales	SUM(sales_transactions[TotalSales])	Total revenue
Total Profit	SUM(sales_transactions[Profit])	Total profit
Profit Margin %	DIVIDE([Total Profit],[Total Sales],0)*100	Profit percentage
Total Orders	COUNTROWS(sales_transactions)	Count of orders
Total Customers	DISTINCTCOUNT([CustomerID])	Unique customers
YoY Growth %	DIVIDE([Total Sales]-[Sales LY],[Sales LY],0)*100	Year over year growth
Return Rate %	Returned orders / Total orders * 100	Return analysis

6. Dashboard Design & Explanation

The dashboard is organized across three pages, each targeting a specific area of business reporting. A consistent color theme, interactive slicers, and drill-through features are applied throughout all pages.

6.1 Page 1 — Executive Summary

- 5 KPI Cards: Total Revenue, Total Profit, Profit Margin %, Total Orders, Total Customers
- Monthly Sales and Profit Trend Line Chart covering 2023–2024
- Sales by Category shown as a Clustered Bar Chart
- Regional Sales Map Visualization broken down by city
- Payment Method Distribution shown as a Pie Chart
- Slicers: Year, Quarter, Category, and Status

6.2 Page 2 — Sales Analysis

- Sales by Channel displayed as a Clustered Column Chart
- Customer Segment breakdown in a Donut Chart
- Top 10 Products Matrix showing Total Sales, Orders, and Profit
- Year-over-Year Growth Comparison visualized as a Line Chart
- Conditional Formatting applied on the Profit column (Red = Loss, Green = High Profit)

6.3 Page 3 — Inventory & Forecast

- Stock Level by Category shown as a Bar Chart
- Sales Forecast Line Chart comparing Actual vs. Forecasted values with Confidence Bounds
- Reorder Level Analysis to flag low-stock categories

6.4 Advanced Features Implemented

Feature	Implementation
Drill-through	Category-level drill-through from Page 1 to Page 2
Tooltips	Custom report page tooltip on bar charts showing Avg Order Value
Bookmarks	Electronics View and All Categories View bookmarks with toggle buttons
Conditional Formatting	Profit column: red for losses, green for high profit
Dynamic Titles	Chart titles update automatically based on the selected year slicer
AI Smart Narrative	Auto-generated insights text displayed on the Executive Summary page
Forecasting	3-month sales forecast with a 95% confidence interval
Row Level Security	City-level RLS roles configured (e.g., Karachi Manager role)

7. Key Business Insights

The following insights emerged from analyzing the data through the Power BI dashboard:

- Electronics is the highest revenue-generating category, accounting for roughly 22% of total sales
- The Online sales channel is growing faster than In-Store, with a noticeable upward trend in 2024
- Q4 (October through December) consistently records the highest sales across all categories, pointing to strong seasonal demand
- The VIP customer segment generates 3x the average order value of Regular customers, despite making up only around 10% of the total customer base
- A return rate of approximately 12% in the Electronics category warrants investigation and corrective action
- Karachi and Lahore are the top-performing store locations, together contributing over 40% of regional sales
- Food & Beverages shows steady growth with low return rates, making it one of the most stable revenue streams
- Sales forecasting indicates a 7–12% growth in Q4 2024 compared to the same period in 2023

8. Conclusion

This Power BI project successfully brings together data from four different sources into a single, interactive, and professional Business Intelligence dashboard. It meets all minimum requirements — four data sources, over 5,000 rows of data, full ETL transformation in Power Query, a star schema data model, DAX measures, and a multi-page interactive dashboard.

The advanced features — including drill-through navigation, bookmarks, conditional formatting, AI-generated narratives, sales forecasting, and Row Level Security — reflect a level of Power BI proficiency that goes well beyond the basics. The Retail Sales and Inventory dashboard provides actionable insights that help business stakeholders make better decisions across sales performance, customer behavior, inventory management, and planning for the future.

The star schema ensures query efficiency, and the DAX measures enable real-time KPI calculation that responds dynamically to slicer selections. Overall, this project reflects the standards and practices of real-world BI consulting work.

9. References & Data Sources

- Microsoft Power BI Documentation: <https://docs.microsoft.com/power-bi>

- Dataset: Custom generated retail data (products, customers, sales, store targets)
- DAX Guide: <https://dax.guide>
- Power Query M Language Reference: <https://docs.microsoft.com/powerquery-m>